

8_Operators_part1

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```
[2]: # List of Arithmetic Operators in Python
# "+" for Addition
# "-" for subtraction
# "*" for multiplication
# "/" for division
# "%" for Modulus to get remainder of the division
# "**" for exponentiation
# "//" for floor division - rounded to nearest whole number
```

```
[4]: a= 10
b= 3
print(a+b) # Addition
print(a-b) # Subtraction
print(a*b) # Multiplication
print(a/b) # Division
print(a%b) # Remainder of the division
print(a**b) # 2**2 = 4, 3**2 = 9
print(a//b) # 10/3=3.33 and floor value of 3.33 is 3
```

```
13
7
30
3.3333333333333335
1
1000
3
```

```
[5]: # List of Comparison Operators in Python
# 1. == (equals to)
# 2. != (Not equals to)
# 3. < (Less than)
# 4. > (Greater than)
# 5. <= (Less than or equals to)
# 6. >= (Greater than or equals to)
```

```
[6]: x=5
y=10
print((x==y))
```

```
print((x!=y))
print((x<y))
print((x>y))
print((x<=y))
print((x>=y))
```

False
True
True
False
True
False

```
[ ]: # Logical Operators in Python
      # 1. and (Logical AND)
      # 2. or (Logical OR)
      # 3. not (Logical NOT)
```

```
[ ]: a=True
      b=False
      print(a and b)
      print(a or b)
      print(not b)
```

False
True
True

```
[9]: x=5
      y=3
      print(x<10 or y > 10)
```

True

```
[11]: # Assignment Operator
      # = represents assignment and it assigns the value on the right to the variable
      # ↪ on the left
      # +=, -=, *=, /=, %= are compound assignment operators
      a=10
      b=3

      c = a    # simple assignment
      b += 5   # Additional assignment and equivalent to b = b+5
      a -= 2   # Subtraction assignment and equivalent to a = a-2
      c *= 3   # Multiplication assignment and equivalent to c = c*3
      b /= 2   # Division assignment and equivalent to b = b/2
      a %= 4   # Modulus assignment and equivalent to a = a%4
      c //= 2  # Floor Division assignment and equivalent to c = C//2
      b **= 2  # Exponentiation assignment and equivalent to b = b**2
```

```
print("a = ",a)
print("b = ",b)
print("c = ",c)
```

```
a = 0
b = 16.0
c = 15
```

```
[14]: # Special Operators
      # is and is not are the identity operators in python
      # They are used to check whether two variables/values are located on the same
      ↪ part of the memory.
```

```
a = 5
b = 5
print( a is b)

list1 = [1,2,3,4,5]
list2 = [1,2,3,4,5]
print(list1 is list2)

name1="Hanu"
name2="Hanu"
print(name1 is name2)
```

```
True
False
True
```

```
[16]: print(a,b)
      print(id(a), id(b))
      print(list1,list2)
      print(id(list1), id(list2))
      print(name1, name2)
      print(id(name1), id(name2))
```

```
5 5
10446536 10446536
[1, 2, 3, 4, 5] [1, 2, 3, 4, 5]
134863210210304 134863188760192
Hanu Hanu
134863188004512 134863188004512
```